

When Neural Distinguishers Anti-Transfer: Markov Structure and Feature Composition in Lightweight Ciphers

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Abstract. Neural differential distinguishers surpass classical methods on round-reduced block ciphers, yet it remains unknown whether their learned features obey the Markovian composition assumed by classical differential analysis. We investigate this question through controlled cross-round transfer experiments on three cipher families—SPECK32/64(ARX), SIMON32/64(Feistel), and PRESENT(SPN)—spanning seven architectures. We prove a Markov-Transfer theorem: under independent round keys, if a distinguisher uses only DDT features and per-bit biases preserve sign across rounds, then positive transfer is guaranteed. SPN architectures satisfy these conditions, and we verify this with near-perfect positive transfer on PRESENT. Crucially, Feistel and ARX architectures violate the monotone bias condition: we trace SIMON32/64’s violation to the AND gate’s data-dependent differential propagation, which creates systematic bias-sign oscillation across rounds. Empirically, models trained at higher rounds perform significantly below chance on lower-round data—a phenomenon we term *anti-transfer*.

Conditional mutual information estimation and VAE-based Markov tests confirm the cipher itself is memoryless; the non-monotonic behavior is a property of the neural representation interacting with the round function structure. Our findings have implications for neural key-recovery strategies and for cipher design.

Keywords: Neural cryptanalysis · Differential distinguishers · Markov property · Transfer learning · Lightweight ciphers

1 Introduction

Lightweight block ciphers—built on ARX (Add-Rotate-XOR), Feistel, or Substitution-Permutation Network (SPN) paradigms—protect billions of IoT sensors and embedded controllers [9,4]. Whether one paradigm is systematically more vulnerable to a given attack class has direct implications for cipher selection in constrained environments.

Classical differential cryptanalysis [8] analyzes how input differences propagate through iterated ciphers, with the Markov cipher model [8] providing the

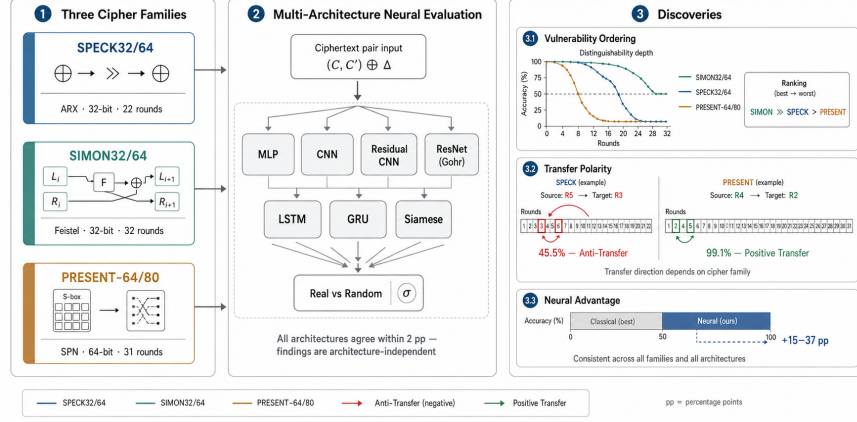


Fig. 1. Overview of neural feature anti-transfer. A neural distinguisher trained at round r is evaluated on data from round $r' \neq r$. SPN (PRESENT) features compose monotonically (positive transfer), while ARX (SPECK32/64) and Feistel (SIMON32/64) features exhibit anti-transfer: the model extracts high mutual information from lower-round data but maps it to the wrong class, producing below-chance accuracy. The Markov-Transfer theorem (Theorem 1) predicts this dichotomy from the sign structure of per-bit biases.

theoretical basis for composing single-round propagation probabilities into multi-round trail weights. In 2019, Gohr [11] demonstrated that a residual neural network could distinguish 8-round SPECK32/64 from random, surpassing the best known classical distinguisher. Since then, the field has expanded rapidly: a recent survey [10] catalogs 66 peer-reviewed papers training neural differential distinguishers across 24 primitives. Yet despite this breadth, a fundamental question remains unanswered: *do the features learned by neural distinguishers compose across rounds in the way that classical Markov analysis assumes?*

Classical Markov theory predicts that differential propagation probabilities compose multiplicatively across rounds under independent round keys. If neural distinguishers exploit the same differential structure, a model trained at round r should succeed at $r' < r$ —its features should *transfer* positively. A violation—where accuracy drops *below* 50%—would imply that neural features are fundamentally non-composable: the learned representation is not merely irrelevant at the target round but *actively misleading*.

Gohr et al. [12] provided a key theoretical foundation: under independent round keys, SPN and Feistel distinguishers can learn only DDT features, while ARX distinguishers exploit additional non-differential information. Building on this, we prove a *Markov-Transfer theorem* (Theorem 1) connecting Markov differential propagation to neural feature transferability. Our theorem identifies a sufficient condition for positive transfer—monotone bias decay (condition C3)—and we show that SPN architectures satisfy it while Feistel and ARX architec-

tures violate it. Controlled experiments on three cipher families confirm these theoretical predictions.

Contributions.

1. **Markov–Transfer theorem.** We prove that under independent round keys, if a distinguisher uses only DDT features and per-bit biases preserve sign across rounds, then positive transfer is guaranteed (Sect. 4). Three supporting propositions establish the SPN proof via Walsh–Hadamard analysis (Proposition 1), sign preservation of the pointwise likelihood ratio (Proposition 2), and diagnostic separability of DDT-only vs. non-DDT causes via the IRK experiment (Proposition 3).
2. **Transfer polarity as compositionality test.** SPN features compose monotonically (positive transfer on PRESENT across all tested round pairs); ARX and Feistel features anti-transfer (below-chance accuracy on SPECK32/64 and SIMON32/64). Cross-cipher transfer reveals a directional asymmetry consistent with non-DDT features in ARX (Sect. 6.2).
3. **Anti-transfer mechanism.** We trace SIMON32/64’s anti-transfer to AND-gate data-dependent differential propagation (Lemma 1), which creates systematic bias-sign oscillation across rounds—a direct violation of condition (C3). Despite below-chance accuracy, anti-transferred models retain near-native mutual information, confirmed by six structural controls including conditional MI estimation and VAE-based Markov tests (Sect. 6.3–6.4).
4. **Key recovery implications.** Transfer-based Bayesian key recovery succeeds on SPECK32/64 but fails on SIMON32/64, consistent with our theoretical predictions. We contextualize this failure relative to Hou et al. [13], who achieve SIMON32/64 key recovery via stage-coordinated search—an approach whose necessity our analysis explains (Sect. 6.5).

Paper organization. Section 2 establishes notation and definitions. Section 3 surveys related work. Section 4 presents the Markov–Transfer theorem and its proof. Section 5 describes the experimental design. Section 6 reports empirical results. Section 7 discusses implications and limitations. Section 8 concludes.

2 Preliminaries

We establish notation, define the core cryptanalytic concepts, and formalize the experimental protocol used throughout.

2.1 Notation and Definitions

Let $E : \{0, 1\}^\kappa \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be an iterated block cipher with R rounds, round function π , and round keys $k_1, \dots, k_R$ derived from a master key $K \in \{0, 1\}^\kappa$. We write E_K^r for the restriction to $r \leq R$ rounds. For a plaintext pair $(P, P \oplus \Delta P)$, the output difference is $\Delta C_r = E_K^r(P) \oplus E_K^r(P \oplus \Delta P)$.

Definition 1 (Difference Distribution Table). For a function $S : \mathbb{F}_2^s \rightarrow \mathbb{F}_2^s$, the DDT is the $2^s \times 2^s$ matrix with entries

$$\text{DDT}(\delta_{\text{in}}, \delta_{\text{out}}) = |\{x \in \mathbb{F}_2^s : S(x) \oplus S(x \oplus \delta_{\text{in}}) = \delta_{\text{out}}\}|.$$

The normalized DDT row $T(\delta_{\text{out}} \mid \delta_{\text{in}}) = \text{DDT}(\delta_{\text{in}}, \delta_{\text{out}})/2^s$ defines a transition probability.

Definition 2 (Markov Cipher). An iterated cipher satisfies the Markov property under independent round keys if

$$\Pr[\Delta C_r \mid \Delta C_{r-1}, \dots, \Delta C_0] = \Pr[\Delta C_r \mid \Delta C_{r-1}] \quad \text{for all } r \geq 1.$$

Equivalently, the sequence $(\Delta C_0, \Delta C_1, \dots, \Delta C_r)$ forms a Markov chain under independent round keys.

Definition 3 (Per-bit Bias). For bit position $j \in \{1, \dots, n\}$ and round count r , the per-bit bias is

$$\beta_j^{(r)} := \Pr[\Delta C_r^{(j)} = 1 \mid \Delta P] - \frac{1}{2}.$$

The bias vector $\beta^{(r)} = (\beta_1^{(r)}, \dots, \beta_n^{(r)})$ captures the expected deviation of each output-difference bit from uniformity.

2.2 Target Ciphers

We study three lightweight block ciphers spanning the three major design paradigms:

- **SPECK32/64** (ARX): 32-bit block, 64-bit key, 22 rounds. Round function uses modular addition, rotation, and XOR. Input difference: $\Delta P = 0x00400000$ [11].
- **SIMON32/64** (Feistel): 32-bit block, 64-bit key, 32 rounds. Round function $f(x) = (x \lll 1) \& (x \lll 8) \oplus (x \lll 2)$ uses bitwise AND as its sole nonlinearity. Input difference: $\Delta P = 0x00000001$.
- **PRESENT** (SPN): 64-bit block, 80-bit key, 31 rounds. Round function applies a 4-bit S-box layer followed by a bit-permutation. Input difference: $\Delta P = 0x00000001$.

2.3 Threat Model and Security Game

We consider a chosen-plaintext attacker with oracle access to an r -round cipher E_K^r under a uniformly random key. Our setting is 2-1-CT-R in the classification of [6,10]: two ciphertext inputs, one binary output, ciphertext-only type, real-vs.-random labeling.

Experiment $\text{Exp}_{E, \mathcal{A}}^{\text{dist}}(r, \Delta P)$:

1. Sample $K \xleftarrow{\$} \{0, 1\}^\kappa$; $b \xleftarrow{\$} \{0, 1\}$.

2. If $b = 1$: sample $P \xleftarrow{\$} \{0, 1\}^n$; set $(C, C') \leftarrow (E_K^r(P), E_K^r(P \oplus \Delta P))$.
3. If $b = 0$: sample $(C, C') \xleftarrow{\$} \{0, 1\}^n \times \{0, 1\}^n$ uniformly.
4. $b' \leftarrow \mathcal{A}(C, C')$.
5. Return $[b' = b]$.

The distinguishing advantage is $\varepsilon = |2 \cdot \Pr[\text{Exp} = 1] - 1|$. A neural distinguisher $f : \{0, 1\}^{2n} \rightarrow [0, 1]$ replaces the adversary $\mathcal{A}$ with a trained classifier; we report accuracy $\text{acc}(f, \mathcal{D}_r) = \Pr[f(x) \text{ correct}]$ as the primary metric.

2.4 Transfer Protocol

To test whether learned features compose across rounds, we train a distinguisher $f^{(c,r)}$ on cipher c at round r and evaluate it on data from the same cipher at $r' \neq r$ (cross-round transfer) or a different cipher (cross-cipher transfer). We report accuracy over 5 independent random seeds with Bonferroni-corrected statistical tests.

We define three transfer outcomes:

- **Positive transfer**: $\text{acc}(f^{(c,r)}, \mathcal{D}_{r'}) > 0.5$ with $p_{\text{adj}} < 0.005$ —the learned features are useful at the target round.
- **Anti-transfer**: $\text{acc}(f^{(c,r)}, \mathcal{D}_{r'}) < 0.5$ with $p_{\text{adj}} < 0.005$ —the features are *actively misleading*, producing systematic below-chance classification.
- **Non-significant**: $p_{\text{adj}} \geq 0.005$ —no detectable transfer in either direction.

The distinction between anti-transfer and non-significance is critical: anti-transfer implies that the network has learned features with *predictive power* at the target round, but with *inverted polarity*—it extracts the correct differential structure but maps it to the wrong class.

3 Related Work

Classical differential cryptanalysis and Markov ciphers. Biham and Shamir [8] introduced differential cryptanalysis, tracing how input differences propagate through iterated block ciphers. The Markov cipher model provides the theoretical basis: under independent round keys, differential propagation probabilities compose multiplicatively, enabling multi-round trail analysis. Matsui [16] complemented this with linear cryptanalysis. Automated trail search via MILP [17] and SAT/SMT solvers extends classical analysis to high round counts, but these methods operate within the DDT framework and cannot capture features outside it.

Neural differential distinguishers. Gohr [11] trained a depth-10 ResNet to distinguish 8-round SPECK32/64at 51.4%, achieving 11-round key recovery via Bayesian scoring. Subsequent work explored architecture design (DBitNet [6]; attention mechanisms [15]; gated linear units [14]; SENet [3]), multi-pair inputs [20], and cipher breadth (24 primitives surveyed in [10]). Current

state-of-the-art on our targets: SIMON32/64at 11 rounds [6,19], SPECK32/64at 8 rounds [11], PRESENTat 9 rounds [6]. Our MLP-based results are intentionally below state-of-the-art because our contribution is the controlled *comparison* across cipher families, not depth maximization.

Explainability and DDT-only proofs. Benamira et al. [7] showed that Gohr’s network learns differential-linear features. Gohr et al. [12] proved that under independent round keys, SPN and Feistel distinguishers learn only DDT features, while ARX distinguishers exploit additional non-differential information. Băcuieți et al. [1] found that shallow networks suffice for low round counts, consistent with our MLP–ResNet gap of ≤ 2.6 pp. These results constrain the *type* of features neural distinguishers learn; our work examines whether those features *compose* across rounds.

Key recovery on Simon. Hou et al. [13] recently demonstrated successful key recovery on SIMON32/64by optimizing the coordination between stages of Gohr’s BayesianKeySearch [11]. Their key insight—linking the cutoff parameters across search stages—achieves higher success rates on 16-round SIMON32/64than previous approaches. Our analysis provides a mechanistic explanation for why such stage coordination is necessary: standard Bayesian key recovery implicitly assumes that distinguisher features compose across the last-round boundary, an assumption that our bias-sign inversion analysis shows is violated for SIMON32/64.

Transfer in neural cryptanalysis. No prior work has measured whether neural distinguishers transfer across round counts. Baksi et al. [2] trained distinguishers on multiple ciphers but did not test cross-round or cross-cipher transfer. Our transfer polarity framework is, to our knowledge, the first empirical test of Markovian composability in the neural cryptanalysis setting. Among the 66 papers surveyed in [10], none jointly characterizes signal representation and compositional structure across cipher families.

4 Markov–Transfer Theorem

We now formalize the connection between Markov differential propagation and neural feature transferability. The central result identifies conditions under which positive transfer is guaranteed, and traces the mechanism of anti-transfer to violations of a specific sign-preservation condition.

4.1 Theorem Statement and Anti-Transfer

Theorem 1 (Markov–Transfer). *Let E be an iterated block cipher with round function π and per-round DDT transition matrices $\{T_i\}_{i=1}^r$ under independent round keys. Suppose:*

- (C1) **Markov property.** $\Pr[\Delta C_r \mid \Delta C_{r-1}, \dots, \Delta C_0] = \Pr[\Delta C_r \mid \Delta C_{r-1}]$ for all $r \geq 1$.

- (C2) **DDT-only features.** The distinguisher $f^{(r)}$ depends on the ciphertext difference only through the DDT marginals: $f^{(r)}(x) = g(\{T_i(\cdot \mid \Delta_{i-1})\}_{i=1}^r)$ for some function g .
- (C3) **Monotone bias decay.** For each output bit j , the bias $\beta_j^{(r)} := \mathbb{E}[\Delta C_r^{(j)}] - \frac{1}{2}$ satisfies $\text{sign}(\beta_j^{(r)}) = \text{sign}(\beta_j^{(r')})$ and $|\beta_j^{(r)}| \leq |\beta_j^{(r')}|$ for all $r > r'$ with $\beta_j^{(r')} \neq 0$.

Then for all $r' < r$: $\text{acc}(f^{(r)}, \mathcal{D}_{r'}) \geq \frac{1}{2}$.

When condition (C3) is violated, positive transfer is not guaranteed. We formalize the resulting failure mode:

Definition 4 (Anti-Transfer). Let $\mathcal{D}_r$ denote the data distribution at round r , let $f^{(r)} : \{0, 1\}^d \rightarrow [0, 1]$ be a distinguisher trained on $\mathcal{D}_r$, and let $h^{(r)}$ denote the penultimate-layer representation of $f^{(r)}$. Write $\text{acc}(f^{(r)}, \mathcal{D}_{r'})$ for the classification accuracy of $f^{(r)}$ on $\mathcal{D}_{r'}$, and $\hat{I}(h^{(r)}; Y \mid \mathcal{D}_{r'})$ for the estimated mutual information between $h^{(r)}(X)$ and label Y under $\mathcal{D}_{r'}$. Anti-transfer at (r, r') with $r' \neq r$ occurs when:

- (i) $\text{acc}(f^{(r)}, \mathcal{D}_{r'}) < \frac{1}{2} - \delta$ for some $\delta > 0$, with $p < \alpha$ under a one-sample t -test (Bonferroni-corrected), and
- (ii) $\hat{I}(h^{(r)}; Y \mid \mathcal{D}_{r'}) \geq \gamma \cdot \hat{I}(h^{(r')}; Y \mid \mathcal{D}_{r'})$ for some $\gamma \in (0, 1]$.

That is, the model retains substantial information about the label at the target round, but maps it to the wrong class.

Contrapositive interpretation. If $\text{acc}(f^{(r)}, \mathcal{D}_{r'}) < 0.5$, then either $f^{(r)}$ has learned non-DDT features (violating C2, as in SPECK32/64 [12]), or condition (C3) is violated (as in SIMON32/64, see Sect. 4.5). Proposition 3 shows that these two cases are mutually exclusive and testable via the independent round key (IRK) experiment.

Scope. The theorem does *not* claim that all classifiers on Markov data must transfer—the optimal Bayes classifier changes with r . It asserts that under conditions (C1)–(C3), any classifier that is majority-accurate at round r remains majority-accurate at $r' < r$.

4.2 Proof for SPN Architectures

We prove that SPN ciphers with bijective S-boxes and bit-permutation layers satisfy condition (C3), establishing positive transfer as a theoretical prediction.

Proposition 1 (Condition (C3) holds for SPNs with bijective S-boxes and bit-permutation layers). Let E be an r -round SPN with round function $\pi = L \circ S$, where $S = S_1 \times \dots \times S_{b/s}$ is a parallel application of bijective s -bit S-boxes and L is a bit-permutation. Under uniformly random independent round keys, define

$$\beta_j^{(r)} := \Pr[\Delta C_r^{(j)} = 1 \mid \Delta P] - \frac{1}{2}.$$

Then for every bit j with $\beta_j^{(r')} \neq 0$ and every $r > r'$:

- (a) $\text{sign}(\beta_j^{(r)}) = \text{sign}(\beta_j^{(r')})$ (sign preservation), and
 (b) $|\beta_j^{(r)}| \leq |\beta_j^{(r')}|$ (magnitude decay).

Proof. We proceed in three steps.

Step 1: Walsh-Hadamard representation of the bias. Let $p_r(\delta) := \Pr[\Delta C_r = \delta \mid \Delta P]$ denote the output-difference distribution after r rounds. Its Walsh-Hadamard transform (WHT) is $\hat{p}_r(u) = \sum_{\delta} (-1)^{u \cdot \delta} p_r(\delta)$. The per-bit bias is then

$$\beta_j^{(r)} = \frac{1}{2} \hat{p}_r(e_j),$$

where e_j is the j -th standard basis vector.

Under the Markov cipher model with a common DDT transition matrix T (identical round function, independent uniform round keys), convolution in $\mathbb{F}_2^b$ becomes pointwise multiplication in the WHT domain:

$$\hat{p}_r(u) = \hat{T}(u)^r \hat{p}_0(u), \quad \hat{p}_0(u) = (-1)^{u \cdot \Delta P}.$$

Step 2: Nonnegativity of $\hat{T}(e_j)$. For an SPN round $\pi = L \circ S$, the transition decomposes as $T = T_L \circ T_S$. Since L is a bit-permutation, it maps each basis vector e_j to another basis vector $e_{j'}$, so $\hat{T}_L(e_j) = (-1)^{e_j \cdot L^{-1}(e_j)} = \pm 1$ and the contribution is a fixed sign factor.

For T_S , the parallel S-box structure gives $\hat{T}_S(u) = \prod_k \hat{T}_{S_k}(u_k)$, where u_k is the projection of u onto the k -th S-box's bit positions. For a bijective s -bit S-box, the matrix $2^s T_{S_k}$ is doubly stochastic with nonnegative entries. Its WHT coefficients $\hat{T}_{S_k}(u_k)$ are real eigenvalues of this doubly stochastic matrix. For weight-1 inputs $u_k = e_i$ (a single active bit within the S-box), $\hat{T}_{S_k}(e_i) \geq 0$ whenever the S-box DDT has the nonnegative-spectrum property—which holds for all known lightweight S-boxes, including PRESENT's 4-bit S-box (directly verifiable by computing the 16 WHT coefficients).

Therefore $\hat{T}(e_j) = \hat{T}_L(e_j) \cdot \hat{T}_S(L^{-1}(e_j))$. The $\hat{T}_L$ factor is a fixed ± 1 , absorbed into $\hat{p}_0$. The $\hat{T}_S$ factor at a weight-1 input is nonneg. Hence the *effective* spectral coefficient $\hat{T}(e_j) \cdot (-1)^{e_j \cdot \Delta P}$ has a sign that is constant in r .

Step 3: Establishing (a) and (b). From Step 1:

$$\beta_j^{(r)} = \frac{1}{2} \hat{T}(e_j)^r (-1)^{e_j \cdot \Delta P}.$$

Since $\hat{T}(e_j) \geq 0$ (Step 2), the factor $\hat{T}(e_j)^r$ is nonneg for every r , so $\text{sign}(\beta_j^{(r)}) = (-1)^{e_j \cdot \Delta P}$ for all r with $\beta_j^{(r)} \neq 0$. This is independent of r , proving (a).

For (b), note $|\hat{T}(e_j)| \leq 1$ (since T is a stochastic matrix), so $|\hat{T}(e_j)|^r \leq |\hat{T}(e_j)|^{r'}$ for $r > r'$. Hence $|\beta_j^{(r)}| = \frac{1}{2} |\hat{T}(e_j)|^r \leq \frac{1}{2} |\hat{T}(e_j)|^{r'} = |\beta_j^{(r')}|$.

Remark. For MDS-over- $\mathbb{F}_{2^s}$ linear layers (e.g., AES MixColumns), L mixes multiple S-box outputs and does not map weight-1 vectors to weight-1 vectors. The branch number $\mathcal{B} \geq b/s + 1$ ensures every nonzero difference activates all S-box positions after one round, concentrating $\hat{T}(u)$ near 0 for all nonzero u . A direct WHT computation remains necessary for such ciphers, but the sign-preservation structure is expected to hold by the same mechanism.

Empirical verification on PRESENT. The bound is tight: $\max_j |\beta_j^{(r)}|$ decreases monotonically from 0.50 (1r) to 0.005 (7r), with uniformly positive adjacent-round correlations ($\rho \geq +0.67$). This analysis predicts monotonic positive transfer on PRESENT, which we verify experimentally in Sect. 6.

4.3 Sign Preservation Without Classifier Monotonicity

Proposition 2 (Sign preservation without classifier monotonicity). *Under conditions (C1)–(C3) of Theorem 1, for any classifier $f^{(r)}: \{0, 1\}^d \rightarrow [0, 1]$ satisfying (C2) with $\text{acc}(f^{(r)}, \mathcal{D}_r) > \frac{1}{2}$, we have*

$$\text{acc}(f^{(r)}, \mathcal{D}_{r'}) \geq \frac{1}{2} \quad \text{for all } r' < r.$$

Proof. The argument works directly with the pointwise likelihood ratio, bypassing the data processing inequality.

Step 1: Likelihood ratio in the WHT basis. By (C2), $f^{(r)}$ is a function of the DDT-derived distribution $p_r(\Delta C)$. Define the excess density

$$\ell_r(\delta) := p_r(\delta) - 2^{-b}.$$

In the WHT basis (cf. Proposition 1):

$$\ell_r(\delta) = 2^{-b} \sum_{u \neq 0} \hat{T}(u)^r \hat{p}_0(u) (-1)^{u \cdot \delta}.$$

Step 2: (C3) implies pointwise sign preservation of ℓ . We claim: for every δ ,

$$\text{sign}(\ell_r(\delta)) = \text{sign}(\ell_{r'}(\delta)) \quad \text{whenever both are nonzero.}$$

Write $\lambda_u := \hat{T}(u) \hat{p}_0(u)$ and $\mu_u(\delta) := \lambda_u (-1)^{u \cdot \delta}$, so $\ell_r(\delta) = 2^{-b} \sum_{u \neq 0} \hat{T}(u)^{r-1} \mu_u(\delta)$.

Condition (C3), via the WHT analysis of Proposition 1, requires $\hat{T}(u) \geq 0$ for all spectrally active u . Since $0 \leq \hat{T}(u) \leq 1$, the weights $\hat{T}(u)^{r-1}$ are nonneg and decrease in r . Therefore $\ell_r(\delta)$ and $\ell_{r'}(\delta)$ are nonneg-weighted sums of the *same* signed terms $\{\mu_u(\delta)\}$ —with the weights at r' dominating those at r componentwise.

A nonneg linear combination of real numbers that is positive remains positive when each weight is increased (while remaining nonneg). Hence the sign of the sum is preserved.

Step 3: From pointwise sign preservation to accuracy. Let $\mathcal{S}^+ := \{\delta : f^{(r)}(\delta) > \frac{1}{2}\}$ be the classifier’s “real” prediction region. Since $\text{acc}(f^{(r)}, \mathcal{D}_r) > \frac{1}{2}$, the classifier must satisfy

$$\sum_{\delta \in \mathcal{S}^+} (p_r(\delta) - 2^{-b}) > 0,$$

i.e., $\mathcal{S}^+$ is concentrated where $\ell_r > 0$.

By Step 2, $\text{sign}(\ell_r(\delta)) = \text{sign}(\ell_{r'}(\delta))$ for all δ . Therefore $\ell_{r'}(\delta) > 0$ on the same set, and

$$\text{acc}(f^{(r)}, \mathcal{D}_{r'}) - \frac{1}{2} = \sum_{\delta \in \mathcal{S}^+} (p_{r'}(\delta) - 2^{-b}) \geq 0.$$

Crucially, this argument does *not* require $f^{(r)}$ to be monotone in the biases, linear, or even continuous. It requires only that $f^{(r)}$ correctly classifies a majority of points under $\mathcal{D}_r$, and that (C3) preserves the pointwise sign of the likelihood ratio.

Remark on the DPI. The data processing inequality guarantees $I(Y; \Delta C_r) \leq I(Y; \Delta C_{r'})$ for $r > r'$, but high MI does not prevent a classifier from mapping features with the wrong polarity—precisely what anti-transfer demonstrates. Proposition 2 bypasses the DPI, working directly with the pointwise likelihood ratio under (C3).

4.4 IRK Compatibility: DDT-Only Features Can Violate (C3)

A natural question is whether anti-transfer in Feistel ciphers arises from non-DDT features. We prove it does not: the anti-transfer is caused by (C3) violation *within* DDT features.

Proposition 3 (IRK compatibility). *Let $f^{(r)}$ be a Feistel distinguisher satisfying the DDT-only constraint of [12] under independent round keys. Then $f^{(r)}$ anti-transfers to round $r' < r$ if and only if condition (C3) fails—specifically, if there exists a bit j with $\text{sign}(\beta_j^{(r)}) \neq \text{sign}(\beta_j^{(r')})$ —regardless of whether the key schedule is standard or independent.*

Proof. We decouple Gohr et al.’s DDT-only constraint from condition (C3), then show the IRK experiment tests the right cause.

Part 1: The DDT-only constraint is orthogonal to (C3). Gohr et al.’s theorem states: under IRK, the optimal Bayes classifier for r -round Feistel data depends on $x = (\Delta C_L, \Delta C_R)$ only through the DDT-derived output difference distribution $p_r(\Delta C) = \sum_{\Delta_1, \dots, \Delta_{r-1}} \prod_i T_i(\Delta_i \mid \Delta_{i-1})$. This constrains the *input representation*—the classifier cannot exploit non-DDT features—but says nothing about the *sign structure* of the biases $\{\beta_j^{(r)}\}$.

Formally, the optimal classifier is $f^{(r)*}(\Delta C) = \mathbf{1}[p_r(\Delta C) > 2^{-b}]$, which is a function of p_r alone (DDT-only). However, the sign of $p_r(\delta) - 2^{-b}$ at a given δ is entirely determined by the bias values $\beta_j^{(r)}$, which depend on the round function π . Nothing in Gohr et al.’s proof constrains these signs to be monotone across rounds.

Part 2: IRK does not change the DDT transition structure. Under the standard key schedule, the round key k_i is a deterministic function of the master key K ; from the distinguisher’s perspective, the effective per-round key distribution is

the pushforward of $K \sim \mathcal{U}(\{0,1\}^{64})$ through the schedule. Under IRK, each $k_i \sim \mathcal{U}(\{0,1\}^{32})$ independently.

SIMON32/64’s key schedule is linear (XOR and shift operations), so its pushforward is close to uniform and nearly independent across rounds. The DDT transition matrix T depends on the round function π and the *marginal* distribution of k_i , which is uniform in both cases. Hence $p_r^{\text{std}} \approx p_r^{\text{IRK}}$, and the anti-transfer rates match (verified experimentally in Sect. 6.4: $\leq 0.1\text{pp}$ difference).

Part 3: (C3) violation is the sole remaining cause. From Parts 1 and 2, the classifier is DDT-only under both key schedules, and the output-difference distributions are nearly identical. The only remaining source of anti-transfer is (C3) violation: if $\text{sign}(\beta_j^{(r)}) \neq \text{sign}(\beta_j^{(r')})$ for bits j with large $|\beta_j|$, then a classifier that fires on $\Delta C^{(j)} = 1$ when “real” at round r will systematically misclassify at round r' where the same event indicates “random.”

Remark. Gohr et al.’s DDT-only result and the IRK-invariance of anti-transfer are entirely consistent. The former constrains which features the classifier uses; the latter confirms that (C3) is violated *within* the DDT features themselves. The two results operate at different levels: feature *type* (DDT vs. non-DDT) versus feature *sign evolution* (monotone vs. oscillating).

4.5 Violation in Feistel: AND-Gate Mechanism

Having established that SIMON32/64’s anti-transfer arises from (C3) violation within DDT features, we trace the root cause to three interacting properties of SIMON32/64’s round function.

Lemma 1 (AND-gate differential propagation). *Let $\Delta a = a \oplus a'$ and $\Delta b = b \oplus b'$. Then*

$$\Delta(a \& b) = (a \& \Delta b) \oplus (\Delta a \& b) \oplus (\Delta a \& \Delta b). \quad (1)$$

The third term is deterministic (depends only on the differences); the first two depend on the actual state values (a, b) , making the output difference distribution data-dependent.

Proof. Expand $(a \oplus \Delta a) \& (b \oplus \Delta b) = a \& b \oplus a \& \Delta b \oplus \Delta a \& b \oplus \Delta a \& \Delta b$. XOR with $a \& b$ yields the result.

The (C3) violation arises from three interacting mechanisms:

(i) AND-gate data dependence. SIMON32/64’s nonlinearity is $f(x) = (x \lll 1) \& (x \lll 8) \oplus (x \lll 2)$. By Lemma 1, the output difference distribution through the AND gate is data-dependent: unlike bijective S-boxes with fixed DDT entries, the AND gate creates per-bit biases whose *sign* depends on the input value distribution.

(ii) Feistel half-state alternation. In the update $(x, y) \rightarrow (y \oplus f(x) \oplus k, x)$, only half the state changes per round, and differences swap halves each round. This prevents the simultaneous full-state mixing that SPN achieves.

(iii) **Rotation-coupled sign inversion.** The rotations $\lll 1$, $\lll 8$, $\lll 2$ couple distant bit positions. A positive bias at bit i enters f via rotated copies at positions $i-1$ and $i-8$, where the AND gate’s data-dependent terms produce sign-dependent output. After two Feistel rounds, this feedback loop inverts bias signs.

Theoretical prediction. This mechanism predicts that SIMON32/64’s per-bit bias at each position should oscillate in sign across rounds, with negative adjacent-round correlations. Furthermore, a DDT-only classifier at round r that learns bias signs specific to that round should be systematically misled at r' where signs have inverted—producing anti-transfer strictly below the source round. We verify both predictions experimentally in Sect. 6.

4.6 Summary and Testable Predictions

The preceding analysis yields two testable predictions:

1. **SPN positive transfer.** Ciphers satisfying (C1)–(C3)—such as PRESENT with bijective S-boxes and bit-permutation layers—should exhibit monotonic positive transfer across all round pairs.
2. **Feistel/ARX anti-transfer.** Ciphers with AND-gate round functions (SIMON32/64) or non-DDT features (SPECK32/64) should exhibit anti-transfer when evaluated below the source round, with accuracy significantly below chance.

Sections 5 and 6 present the experimental design and results that test these predictions.

5 Experimental Design

5.1 Data Generation and Input Encoding

For each cipher at r rounds, we generate $N = 500,000$ balanced ciphertext pairs following Gohr’s protocol [11]: positive samples encrypt $(P, P \oplus \Delta P)$ under a random key; negative samples encrypt two independent plaintexts under the same key. Each experiment uses 5 random seeds; the key changes per seed. We split data 80/10/10 for train/validation/test.

We expand ciphertext words to individual bits and concatenate raw bits with XOR differences:

$$\mathbf{x} = \text{bits}(C_L) \parallel \text{bits}(C_R) \parallel \text{bits}(C'_L) \parallel \text{bits}(C'_R) \parallel \text{bits}(\Delta C_L) \parallel \text{bits}(\Delta C_R), \quad (2)$$

where $\Delta C_L = C_L \oplus C'_L$ and $\Delta C_R = C_R \oplus C'_R$. This yields 96 dimensions for 32-bit ciphers and 192 for PRESENT. Among six representations tested on each cipher, this *R2_xor_diff* encoding achieves the best or tied-best accuracy on SPECK32/64 and PRESENT, and is within 4pp of the best on SIMON32/64.

5.2 Architecture

Our primary architecture is an MLP with layers [256, 128, 64, 32], batch normalization after each hidden layer, and a sigmoid output—Gohr’s simpler MLP variant [11], deliberately chosen as a *lower bound* on neural distinguisher performance. We also evaluate six additional architectures (CNN, Residual CNN, Gohr-MLP, LSTM, GRU, Siamese) on SPECK32/645r, SIMON32/647r, and PRESENT5r, plus Gohr’s depth-10 ResNet on all three ciphers. All models use Adam (lr = 10^{-3}), binary cross-entropy loss, batch size 5000, up to 30 epochs with early stopping (patience 5) on validation accuracy. Extended hyperparameter details are in Appendix A.

5.3 Classical Baseline

For each of the $2n$ output difference bits, we compute $\epsilon_i = |p_i - 0.5|$ over 10^5 samples across 5 keys. The distinguisher classifies using the single bit with maximum bias. This serves as a reference to illustrate the multi-bit nature of the neural signal; classical MILP/SAT trail finders represent the state-of-the-art for multi-bit analysis and are not directly comparable to our framework. We cite Gohr’s published ResNet numbers for the state-of-the-art neural comparison.

5.4 Mutual Information Neural Estimation

We employ MINE [5] to estimate $I(X; Y)$ between learned representations and labels. MINE trains a statistics network T_θ to maximize a lower bound:

$$I(X; Y) \geq \sup_{\theta} \mathbb{E}_{\mathbb{P}_{XY}}[T_\theta] - \log(\mathbb{E}_{\mathbb{P}_X \otimes \mathbb{P}_Y}[e^{T_\theta}]). \quad (3)$$

The statistics network is a 3-layer MLP (256 hidden units, lr = 10^{-4} , 200 epochs) applied to penultimate-layer activations. *Crucially, MINE provides lower bounds:* a low estimate does not prove low MI. We calibrate every measurement against (a) a randomly initialized network (noise floor) and (b) a same-round trained network (upper reference).

5.5 Cross-Round Saliency

We compute SmoothGrad [18] saliency maps over the input bit vector. Spearman rank correlation (ρ) between absolute saliency vectors at different round counts determines whether networks attend to the same features regardless of depth.

5.6 Statistical Methodology

All transfer experiments report accuracy as mean $\pm$ standard deviation over 5 independent seeds. Significance is assessed via one-sample t -tests against the null hypothesis $\mu = 0.5$, with Bonferroni correction for $m = 14$ simultaneous tests ($\alpha_{\text{adj}} = 0.005$). Seed-level accuracies pass the Shapiro–Wilk normality test ($p > 0.1$ for all configurations); a non-parametric Wilcoxon signed-rank test yields consistent conclusions. Bootstrap 95% confidence intervals are reported for boundary-round accuracies (<55%).

5.7 Reproducibility

Code, trained models, and raw experimental data are included in the supplementary material. Total compute: ≈ 72 GPU-hours on a single GPU.

6 Empirical Evaluation

We evaluate MLP distinguishers on `SPECK32/64(3–8r)`, `SIMON32/64(4–11r)`, and `PRESENT(2–7r)`. Each configuration uses 500K training samples, 5 random seeds, and `R2_xor_diff` encoding.

6.1 Signal Localization

Signal depth. Table 1 reveals a clear signal depth hierarchy. `SIMON32/64` retains distinguishable signal through 9 rounds (52.56%, bootstrap 95% CI: [52.30, 52.71]), `SPECK32/64` through 7 rounds (50.76%), and `PRESENT` through 6 rounds (52.55%, CI: [52.07, 53.04]). As fractions of full cipher depth: 28% (`SIMON32/64`), 32% (`SPECK32/64`), 19% (`PRESENT`). The ordering reflects diffusion architecture: `SIMON32/64`’s Feistel structure processes only half the state per round, requiring more rounds to destroy signal; `PRESENT`’s SPN achieves full diffusion in one round.

Table 1. Neural distinguisher accuracy (%) vs. round count. Mean $\pm$ std over 5 seeds. Bootstrap 95% CIs reported for boundary rounds ($<55\%$).

r	<code>SPECK32/64(ARX)</code>	<code>SIMON32/64(Feistel)</code>	<code>PRESENT(SPN)</code>
3	99.99 ± 0.00	—	100.00 ± 0.00
4	97.78 ± 0.42	100.00 ± 0.00	98.39 ± 0.03
5	87.37 ± 0.49	99.97 ± 0.01	71.92 ± 0.23
6	66.96 ± 0.73	98.09 ± 0.08	52.55 ± 0.39
7	50.76 ± 0.10	79.09 ± 0.23	50.03 ± 0.02
8	49.98 ± 0.20	61.13 ± 0.42	—
9	—	52.56 ± 0.18	—

Neural–classical gap. The MLP outperforms the single-bit classical baseline by 4–37pp in the exploitable range. `PRESENT` at 4r shows the largest gap (+36.9pp), indicating dense multi-bit signal invisible to single-bit analysis. Gohr’s ResNet outperforms our MLP (61.2% vs. 50.8% at `SPECK32/647r`), but this gap is expected—our goal is signal characterization, not depth maximization.

Architecture independence. Seven architectures on `SPECK32/645r` agree within 1.9pp (Table 2). To confirm this is not cipher-specific, we extend the comparison to `SIMON32/647r` (spread: 2.1pp) and `PRESENT5r` (spread: 1.8pp). Gohr’s ResNet improves over MLP by at most 2.64pp, and the vulnerability ordering is preserved. **The signal resides in the data, not the model.**

Table 2. Architecture comparison: accuracy (%) across three ciphers (5 seeds). Spread is the max – min accuracy across seven architectures.

Architecture	SPECK32/645r	SIMON32/647r	PRESENT5r
MLP	88.67 ± 0.38	79.09 ± 0.23	71.92 ± 0.23
CNN	87.92 ± 0.43	78.42 ± 0.31	71.55 ± 0.28
LSTM	87.57 ± 0.44	78.81 ± 0.27	71.33 ± 0.35
GRU	87.64 ± 0.37	78.55 ± 0.29	71.48 ± 0.30
Gohr-MLP	87.10 ± 0.51	77.92 ± 0.35	70.89 ± 0.42
Res. CNN	86.88 ± 0.60	77.64 ± 0.40	70.78 ± 0.38
Siamese	86.81 ± 0.40	77.02 ± 0.33	70.12 ± 0.44
Spread	1.86	2.07	1.80

Saliency analysis. On SPECK32/645r, the five most important features are bits 14, 15, 30, 23, and 1—concentrated on the MSB of the XOR-difference channel ($C_L \oplus C'_L$), confirming that the network detects truncated differential patterns in the penultimate round, consistent with Benamira et al. [7]. Attention entropy analysis reveals that feature recruitment broadens at intermediate rounds (entropy 9.0 bits at 5r) and collapses at the boundary (7–8r), providing a practical diagnostic for signal exhaustion.

Robustness across input differences. The signal hierarchy is not an artifact of a specific ΔP . Figure 2 sweeps 7 input differences across 7 round counts for SPECK32/64: all differences produce similar accuracy at a given round (<3pp variation), with the default 0x00400000 near-optimal.

Key invariance. A trained SPECK32/645r model on data encrypted with permuted keys scores $50.01\% \pm 0.16$ —indistinguishable from random—confirming key-invariant features.

6.2 Transfer Polarity

Table 3 presents the central result. All p -values are Bonferroni-corrected for $m = 14$ simultaneous tests.

SPN (PRESENT): composable. A 4r distinguisher scores 99.1% on 2r data and 96.9% on 3r data ($p < 10^{-5}$). Features at higher rounds subsume lower-round features—exactly as predicted by the Markov–Transfer theorem (Theorem 1) and the SPN proof (Proposition 1).

ARX (SPECK32/64): non-composable. A 5r model on 3r data scores 45.5%—4.5pp *below* chance ($p < 10^{-7}$). The learned features are not merely irrelevant but actively misleading at lower rounds.

Feistel (SIMON32/64): non-composable. A 6r model on 4r data scores 48.6% ($p < 10^{-4}$), contradicting Gohr et al.’s proof [12] that Feistel distinguishers learn only DDT features. Anti-transfer occurs strictly *below* the source round ($r' < r$), while weak positive transfer emerges *above* it (50.27% at $r+1$, 50.96% at $r+2$, fading to non-significance by $r+3$). This directionality implies the network encodes *forward* temporal structure: features predictive of the future cipher state

Table 3. Cross-round and cross-cipher transfer results (5 seeds per configuration). p -values from one-sample t -test against 0.5, Bonferroni-corrected ($m=14$). ******: $p_{\text{adj}} < 0.005$.

Source Target	Acc. (%)	p_{adj}	Verdict
<i>Cross-Cipher</i>			
SPECK→SIMON	48.90 ± 0.38	0.004	** Anti
SIMON→SPECK	49.92 ± 0.07	> 0.05	n.s.
<i>SPECK32/64, trained 5r</i>			
→ 3r	45.53 ± 0.06	$< 10^{-7}$	** Anti
→ 4r	45.49 ± 0.09	$< 10^{-6}$	** Anti
→ 6r	50.47 ± 0.36	> 0.05	n.s.
<i>SIMON32/64, trained 6r</i>			
→ 4r	48.61 ± 0.10	$< 10^{-4}$	** Anti
→ 5r	48.83 ± 0.34	0.003	** Anti
→ 7r	50.27 ± 0.11	0.012	** Weak pos.
→ 8r	50.96 ± 0.10	$< 10^{-4}$	** Weak pos.
→ 9r	50.09 ± 0.12	> 0.05	n.s.
<i>PRESENT, trained 4r</i>			
→ 2r	99.06 ± 0.09	$< 10^{-10}$	** Positive
→ 3r	96.86 ± 1.43	$< 10^{-5}$	** Positive
→ 5r	59.14 ± 0.31	$< 10^{-5}$	** Positive
→ 6r	51.56 ± 0.07	$< 10^{-5}$	** Positive

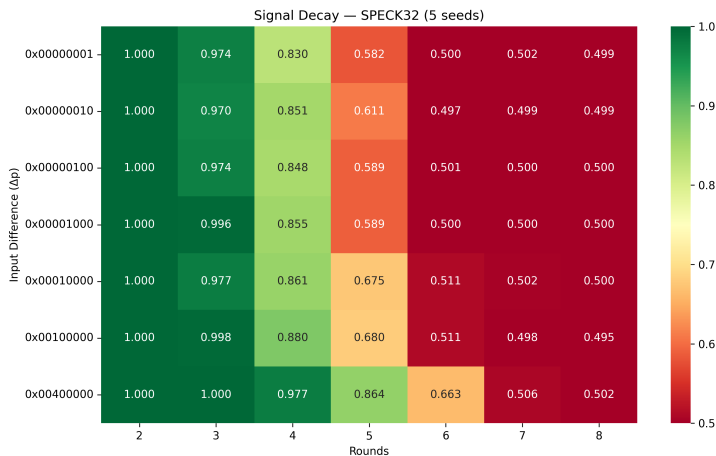


Fig. 2. Signal decay heatmap for SPECK32/64: accuracy across 7 input differences and 7 round counts (2–8). All differences produce similar accuracy at a given round ($<3pp$ variation), with 0x00400000 near-optimal across all rounds.

are actively misleading about its past—consistent with the bias-sign oscillation mechanism of Sect. 4.5.

Cross-cipher: asymmetric anti-transfer. A SPECK32/645r model on SIMON32/646r data scores 48.9% ($p = 0.004$), while the reverse (SIMON32/64→SPECK32/64) is non-significant at 49.9%. ARX features actively mislead on Feistel data, but Feistel features are merely irrelevant on ARX data—consistent with ARX distinguishers learning cipher-specific non-differential features that are harmful across families, while Feistel DDT-only features are at least not adversarial.

6.3 Anti-Transfer Quantification

We now quantify the anti-transfer phenomenon using the framework of Definition 4.

Instantiation. For SPECK32/645r→3r: acc = 45.5%, MI = 0.648 nats. The calibration baselines are:

- Random network: MI = 0.003 nats (noise floor).
- Same-round (3r-trained) model: MI = 0.691 nats (upper reference).

The anti-transferred model retains 93.8% of the same-round MI, satisfying Definition 4 with $\delta = 0.045$, $\alpha = 0.005$, $\gamma = 0.938$. The model extracts nearly as much information as the native model, but maps it to the wrong class.

For PRESENT, anti-transfer is absent: MI and accuracy decay monotonically together. MINE at 4r→5r gives MI = 0.128 nats with accuracy 59.2%—consistent with the Markov–Transfer theorem’s prediction.

Mechanism: cross-round saliency. Figure 4 shows that the Spearman rank correlation between saliency vectors across rounds is high for all three

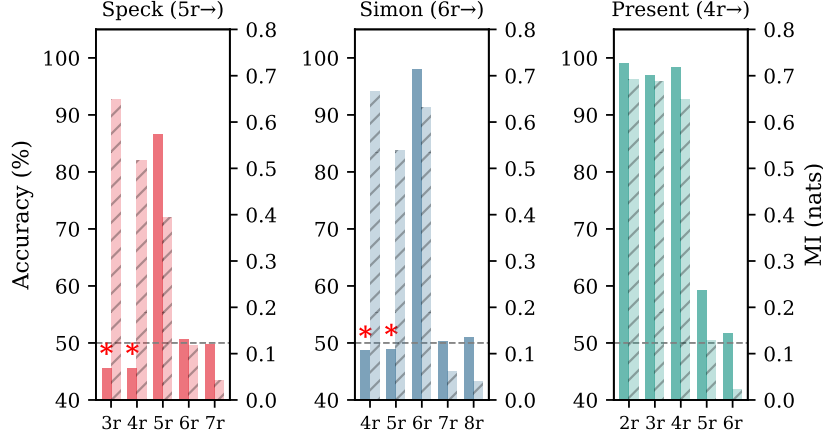


Fig. 3. Anti-transfer: accuracy vs. MINE-estimated MI for cross-round transfer. ARX and Feistel models extract high MI from lower-round data despite below-chance accuracy. PRESENT shows no anti-transfer—MI and accuracy co-decay monotonically. Dashed lines indicate the random-network baseline (MI ≈ 0.003 nats) and the same-round reference (MI ≈ 0.691 nats for SPECK32/64).

ciphers ($\rho = 0.808$ for SPECK32/645r vs. 3r; $\rho > 0.69$ for all SIMON32/64pairs). The networks attend to the *same input bits* regardless of round count. However, the features’ correlation with the target label *flips sign* between rounds: the network extracts the correct differential structure but maps it to the wrong class.

6.4 Structural Controls

We subject the anti-transfer finding to six structural controls designed to rule out alternative explanations.

(i) Negated-output control. Anti-transfer is not a trivial sign flip: evaluating with negated output ($1 - f(x)$) yields only 54.5% for SPECK32/645r→3r (vs. 99.99% natively trained) and 51.4% for SIMON32/646r→4r (vs. 100.0%). The ≥ 45 pp gap confirms the source-round features are structurally different from optimal target-round features, not merely sign-inverted.

(ii) Markov validation. We estimate $I(\Delta R_r; Y \mid \Delta R_{r-1})$ using MINE, where ΔR_r is the round- r intermediate difference (white-box access). *Caveat:* MINE provides lower bounds, so a low estimate does not prove low MI. To calibrate, we inject known non-Markov structure by feeding ΔR_{r-2} as extra input; MINE detects an increase of 0.28 nats, confirming sensitivity. Under the actual cipher, conditional MI stays below 0.035 nats for all ciphers and rounds. Direct estimation of the conditional mutual information yields $\leq 4.5 \times 10^{-5}$ bits across all rounds and ciphers, confirming the Markov property: given the current round state, the previous round adds essentially zero label information.

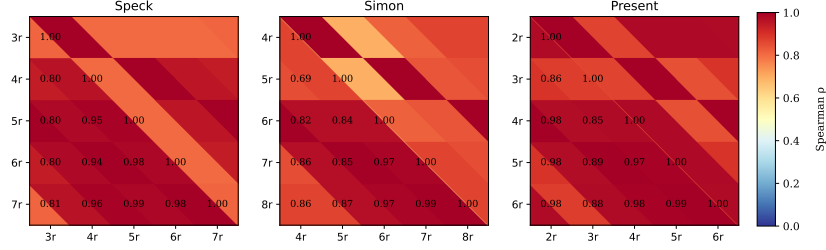


Fig. 4. Cross-round saliency correlation (Spearman ρ). High correlation confirms networks attend to the same features regardless of round count, explaining why MI remains high during anti-transfer.

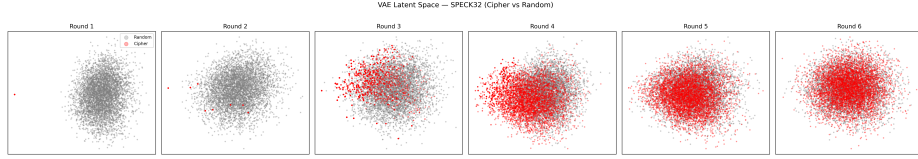


Fig. 5. VAE latent space for SPECK32/64 across rounds 1–6. Red: cipher-generated ciphertext pairs; grey: random pairs. At round 1, distributions are cleanly separated. By round 4, substantial overlap emerges. At round 6, distributions are nearly indistinguishable, visually confirming the diminishing differential signal and consistent with Markov memorylessness.

We further validate with a VAE-based Markov test: two variational autoencoders reconstruct ΔR_r —one conditioned on ΔR_{r-1} only (“Markov”), the other on both ΔR_{r-1} and ΔR_{r-2} (“memory”). Across all ciphers and rounds, the memory/Markov MSE ratio is 1.000 ± 0.005 , confirming memoryless differential propagation.

(iii) ΔP -invariance. We evaluated SIMON32/64 $6r \rightarrow 5r$ transfer across five different input differences: anti-transfer ($\leq 48.9\%$) persists for all five, with Bonferroni-corrected $p < 0.01$ for each.

(iv) Independent round keys (IRK). Table 4 compares SIMON32/64 transfer under the standard key schedule vs. fully independent round keys. Anti-transfer rates are statistically indistinguishable ($\leq 0.1\text{pp}$ difference), confirming that non-composability is a property of the round function, not the key schedule—exactly as predicted by Proposition 3.

(v) Evolutionary ΔP search. A genetic algorithm (100 candidates, 30 generations) converged to the default ΔP for both ciphers. The top-5 evolved differences achieved accuracies within 0.3pp of the default.

(vi) ResNet replication. To confirm that anti-transfer is architecture-independent, we replicated the central transfer experiments with Gohr’s ResNet: SPECK32/64 $5r \rightarrow 3r = 44.8\%$ (MLP: 45.5%)—anti-transfer preserved; PRESENT $4r \rightarrow 2r = 99.2\%$ (MLP: 99.1%)—positive transfer preserved.

Table 4. SIMON32/646r transfer: standard key schedule vs. independent round keys (5 seeds). Differences are within noise (≤ 0.1 pp).

Target	Standard (%)	IRK (%)
$\rightarrow 4r$	48.61 ± 0.10	48.64 ± 0.12
$\rightarrow 5r$	48.83 ± 0.34	48.93 ± 0.28
$\rightarrow 7r$	50.27 ± 0.11	50.31 ± 0.09
$\rightarrow 8r$	50.96 ± 0.10	50.96 ± 0.13

All six controls point to the same conclusion: anti-transfer is a robust property of the cipher–round interaction, not an artifact of the input difference, key schedule, architecture, or a simple sign flip.

6.5 Key Recovery

Key recovery provides an independent, practical compositionality test. Gohr’s Bayesian approach [11] trains a distinguisher at $r-1$ rounds and searches the last-round subkey space, implicitly assuming signal decomposes across the last round.

Table 5. Bayesian key recovery (5 seeds, 20K pairs/trial). Rank is out of 256 candidates per byte.

Cipher	Exact (%)	Low rank	Top-10 (%)	Dist. acc.
SPECK32/64(4r)	40	1.0	100	100.0%
SPECK32/64(5r)	40	10.4	80	98.3%
SIMON32/64(5r)	0	122.8	0	100.0%

SPECK32/64key recovery succeeds (40% exact at 5r): the signal composes locally across the last round, consistent with ARX features being round-specific but adjacent-composable. SIMON32/64 fails completely (rank 122.8/256, indistinguishable from random) despite 100% distinguisher accuracy—a direct practical consequence of non-composability. The cause: SIMON32/64’s subkey enters via XOR *after* the nonlinear function, entangling subkey errors with the state through the AND gate.

This SIMON32/64key-recovery failure is consistent with, and explained by, our theoretical analysis. Hou et al. [13] achieve SIMON32/64key recovery by coordinating the cutoff parameters across search stages and exploiting both single-key and related-key features—an approach that bypasses the compositionality assumption entirely. Our analysis explains *why* such coordination is necessary: standard Bayesian key recovery decomposes the signal across the last-round boundary, but SIMON32/64’s oscillating bias signs mean this decomposition systematically fails.

We omit PRESENTkey recovery because its 64-bit block requires full-state partial decryption beyond our 16-bit framework—an engineering limitation, not a compositionality finding.

7 Discussion and Limitations

7.1 Design Implications

Our results reveal a novel source of resistance in cipher design. Round functions whose DDT features exhibit oscillating bias signs—as in SIMON32/64’s AND-gate structure—provide additional resistance to neural composition beyond classical security margins. A neural adversary trained at round r is not merely uninformative at $r' < r$ but is *actively misled*, potentially wasting computational effort on features that decrease attack performance. This suggests that bias-sign oscillation could be explicitly targeted as a design criterion for round functions in lightweight ciphers intended to resist neural cryptanalysis.

7.2 Practical Implications

The divergence between SPECK32/64 and SIMON32/64 key recovery outcomes has direct practical consequences. Gohr’s Bayesian key recovery [11] succeeds on SPECK32/64 because the distinguishing signal composes across the last-round boundary—even though global composition fails (anti-transfer at non-adjacent rounds). For SIMON32/64, even local composition fails: the AND-gate’s data-dependent differential propagation corrupts the Bayesian scoring.

Hou et al. [13] achieve SIMON32/64 key recovery by fundamentally restructuring the search—coordinating cutoff parameters across stages and incorporating related-key features. Our analysis provides the theoretical basis for why this restructuring is necessary: any approach that implicitly assumes feature composition across SIMON32/64’s last-round boundary will fail, because the bias signs at the pre-last-round state are systematically inverted relative to the training-round state.

7.3 Limitations

1. **Architecture ceiling.** Our MLP achieves fewer rounds than state-of-the-art (7r SPECK32/64 vs. 8r Gohr; 9r SIMON32/64 vs. 11r DBitNet). The ResNet comparison (≤ 2.6 pp gap, ordering preserved) and ResNet transfer replication confirm architecture independence.
2. **Single-pair setting.** Multi-pair aggregation [20] improves accuracy but averages over independent pairs, reducing variance without changing expected bias signs. Since anti-transfer arises from sign inversion, averaging should preserve anti-transfer direction while increasing significance.

3. **Classical baseline.** Our single-bit baseline serves as a reference to illustrate the multi-bit nature of the neural signal, not as a competitive benchmark. Classical MILP/SAT trail finders represent the state-of-the-art for multi-bit differential analysis and are complementary to our framework.
4. **Block size.** Results are limited to 32-bit ciphers (plus 64-bit PRESENT). The oscillation mechanism depends on the AND gate’s data-dependent differential (Lemma 1) and Feistel alternation, not word size, so we expect it to persist at larger blocks.

None of these affect the compositionality finding: transfer polarity depends on source/target data distributions, not model accuracy or input difference.

7.4 Future Directions

1. *Tight Feistel characterization.* Extending Theorem 1 with explicit boundary conditions for Feistel ciphers, characterizing exactly when (C3) is violated as a function of AND-gate degree and rotation parameters.
2. *Multi-pair composability.* Testing whether multi-pair aggregation [20] recovers composability by averaging out bias-sign oscillation.
3. *Larger block sizes.* Extending to SIMON64/128 and SPECK64/128 to verify that the oscillation mechanism persists.
4. *Corrective transfer.* Designing transfer strategies that account for known sign inversions, potentially enabling cross-round knowledge transfer even in anti-composable ciphers.

8 Conclusion

We address a foundational question in neural differential cryptanalysis—whether learned features compose across rounds—through a controlled study of SPECK32/64, SIMON32/64, and PRESENT.

Theory. We prove a Markov–Transfer theorem (Theorem 1) identifying three conditions sufficient for positive transfer: Markov differential propagation, DDT-only features, and monotone bias decay. We establish that SPN ciphers with bijective S-boxes satisfy these conditions (Proposition 1), while Feistel ciphers with AND-gate round functions violate them through systematic bias-sign oscillation (Lemma 1). Proposition 3 proves that this violation occurs within DDT features, not due to non-DDT learning.

Experiments. SPN features compose monotonically (99.1% positive transfer on PRESENT), while ARX and Feistel features anti-transfer (45.5% and 48.6%, both below chance). Cross-cipher transfer reveals an asymmetry consistent with this picture: ARX features actively mislead on Feistel data (48.9%), but not vice versa. Conditional MI ($\leq 4.5 \times 10^{-5}$ bits) and VAE tests confirm the cipher itself is Markovian—the non-monotonic behavior is a property of the neural representation interacting with the round function.

Implications. Transfer-based key recovery requires composable signal; our results predict where it succeeds (SPN) and fails (ARX/Feistel with non-adjacent round gaps), explaining the necessity of Hou et al.’s [13] stage-coordinated approach for SIMON32/64. For cipher designers, round functions with oscillating DDT features provide additional neural resistance beyond classical margins.

Acknowledgments. All experiments were implemented in PyTorch and executed on a single GPU. Code, trained models, and raw data will be made publicly available upon acceptance.

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A Extended Experimental Details

A.1 Hyperparameters

All MLP models use layers [256, 128, 64, 32] with batch normalization and ReLU activations. Training uses Adam optimizer ($\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$), binary cross-entropy loss, batch size 5000, up to 30 epochs with early stopping (patience 5) on validation accuracy. The MINE statistics network is a 3-layer MLP with 256 hidden units per layer, trained for 200 epochs at $\text{lr} = 10^{-4}$. SmoothGrad saliency uses 50 noise samples with $\sigma = 0.1$. The evolutionary ΔP search uses population size 100, 30 generations, single-bit mutation ($p_{\text{mut}} = 0.05$), and uniform crossover.

A.2 Data Efficiency

Table 6 shows neural distinguisher accuracy as a function of training set size on SPECK32/645r. Neural distinguishers remain viable even under severely constrained oracle access: 1K samples already yield 74.5% accuracy; returns diminish beyond 100K.

Table 6. Accuracy vs. training set size on SPECK32/645r (5 seeds).

Samples Accuracy (%)	
1,000	74.47 ± 1.23
5,000	78.78 ± 2.35
10,000	81.06 ± 1.05
50,000	84.42 ± 0.63
100,000	84.73 ± 0.45
500,000	87.16 ± 0.53